
SUMMARY

Software engineer with experience building full-stack applications, REST APIs, database-backed systems, reusable frontend components, and real-time technical projects using Python, JavaScript, TypeScript, SQL, FastAPI, Node.js, Express, React, PostgreSQL, and Docker. Strong foundation in API design, relational data modeling, authentication flows, data validation, debugging, Git-based development, and Agile collaboration. Built projects involving a 26-endpoint FastAPI/PostgreSQL API, OAuth-based Spotify application, React/TypeScript degree planning platform, Dockerized environments, and real-time sensor-data inference.

EDUCATION

University of Massachusetts Amherst | College of Information and Computer Sciences

Bachelor of Science in Computer Science, May 2025 | GPA 3.408 (Spring 2025 term GPA 3.936)

Relevant coursework: Software Engineering, Operating Systems, Computer Networks, Algorithms, Machine Learning, Statistics, Linear Algebra, Computer Systems Principles, Programming with Data Structures, Reasoning Under Uncertainty

PROJECTS

Longitudinal Lesion Tracking REST API | Python, FastAPI, SQLAlchemy, PostgreSQL, Docker Compose

- Built a Dockerized PostgreSQL-backed REST API with 26 endpoints supporting CRUD operations, nested read views, and analytics-style queries for structured longitudinal measurement records.
- Designed a normalized relational schema with foreign keys, cascading deletes, scoped uniqueness constraints, and validation checks to prevent duplicate or inconsistent records.
- Created FastAPI/OpenAPI documentation and automated demo data workflows to support repeatable testing, endpoint validation, and handoffs.

Spotify Playlist Generator | Node.js, JavaScript, Express, React, Spotify API

- Built a full-stack playlist generation app using Spotify OAuth Authorization Code Flow, session-based authentication, Express backend routes, and a React frontend.
- Implemented backend REST endpoints for track search, audio-feature retrieval, recommendation generation, and playlist creation directly in a user's Spotify account.
- Refactored the frontend into 8 reusable React components, including a drag-and-drop selection flow, while improving route structure, state management, and error handling.

CICS Degree Planner | React, TypeScript, HTML/CSS, API, Docker

- Developed React and TypeScript components for a multi-semester degree planning platform with course selection, schedule visualization, and interactive planning workflows.
- Integrated structured course data into frontend state and reusable UI components while maintaining responsive behavior during frequent user interactions and state changes.
- Collaborated on feature design, debugging, code integration, and iterative delivery within a 6-person Agile team using Docker-based development environments.

Gesture-Controlled Drone System | Python, SVM, ESP32, IMU, Tello SDK

- Built a real-time sensor-driven control system that streamed IMU data from an ESP32, transformed accelerometer and gyroscope signals into FFT-based features, and triggered Tello drone maneuvers.
- Collected and labeled 420 IMU sequences across 7 gestures and 3 users, trained an RBF SVM achieving 94% multi-class accuracy.
- Implemented and documented a low-latency inference pipeline connecting sensor data to gesture classification and drone actions including movement, rotation, and flips.

SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Java, HTML/CSS

Backend and APIs: FastAPI, Node.js, Express, REST APIs, OpenAPI/Swagger, JSON, OAuth, session-based authentication

Frontend: React, reusable components, state management, responsive UI, drag-and-drop interfaces

Databases: PostgreSQL, SQLAlchemy, relational modeling, joins, aggregations, constraints, validation rules

ML and data: scikit-learn, NumPy, pandas, Matplotlib, feature engineering, cross-validation, model evaluation

Tools: Git/GitHub, Linux, Docker, Docker Compose, Agile/Scrum fundamentals

EXPERIENCE

Patient Care Technician - Hematology/Oncology (Boston, MA) | Beth Israel Deaconess Medical Center

April 2026 - Present

- Monitor, document, and validate patient clinical data in Epic EHR workflows to support patient status tracking, care continuity, and escalation.
- Review patient vitals and condition trends in a high-acuity hematology/oncology environment, identifying abnormal changes and communicating findings to nursing staff.

IT Support Administrator (Boston, MA) | Pope John Paul II Polish Language School

September 2017 - August 2023

- Provided onsite technical support for instructors and staff by troubleshooting classroom computers, Wi-Fi connectivity, display systems, printers, AV equipment, and setup issues during live weekend school operations.
- Supported student onboarding, digital records, and staff technology access workflows for a Saturday language school serving 130+ students.